

Trigonometric functions and equations - Set 2 - Solutions

Specification MM4.3-MM4.6.

Answer key

Question	Answer	Topic
1	E	Trigonometric functions and equations
2	E	Trigonometric functions and equations
3	E	Trigonometric functions and equations
4	F	Trigonometric functions and equations
5	D	Trigonometric functions and equations
6	F	Trigonometric functions and equations
7	A	Trigonometric functions and equations

Worked solutions

Question 1 X2-07 · Trigonometric functions and equations · spec MM4.6

Consider the equation $\sin x = k$ where k is a real constant. Which of the following statements is/are true?

- I For every k with $-1 < k < 1$ the equation has exactly two solutions in $0 \leq x < 2\pi$.
- II For $k = 1$ the equation has exactly one solution in $0 \leq x < 2\pi$.
- III For every real k the equation has at least one solution.

Answer: E

I and II are read from the graph of $\sin$ over one period. III is false: $\sin x = 2$ has no solutions.

Question 2 TRG-02 · Trigonometric functions and equations · spec MM4.5

Solve $2\cos^2 x + 3\sin x - 3 = 0$ for $0^\circ \leq x \leq 360^\circ$. How many solutions are there?

Answer: E

$2(1 - \sin^2 x) + 3\sin x - 3 = 0 \Rightarrow 2\sin^2 x - 3\sin x + 1 = 0 \Rightarrow (2\sin x - 1)(\sin x - 1) = 0$. $\sin x = \frac{1}{2}$: $30^\circ, 150^\circ$; $\sin x = 1$: 90° . Three solutions.

Question 3 TRG-05 · Trigonometric functions and equations · spec MM4.4

Which of the following statements about the graph of $y = \cos x$ is/are true?

- I It is symmetric about the y-axis.
- II It is a translation of the graph of $y = \sin x$ by $\frac{\pi}{2}$ in the negative x -direction.
- III It has period π .

Answer: E

$\cos(-x) = \cos x$ so I is true. $\sin(x + \frac{\pi}{2}) = \cos x$, so II is true. The period of $\cos$ is 2π , so III is false.

Question 4 TRG-14 · Trigonometric functions and equations · spec MM4.6

Solve $12\cos^2 x + 6\sin x - 10 = 2$ for $0^\circ < x < 360^\circ$. Which of the following is the complete set of solutions?

Answer: F

$12(1 - \sin^2 x) + 6\sin x - 10 = 2 \Rightarrow 12\sin^2 x - 6\sin x = 0 \Rightarrow 6\sin x(2\sin x - 1) = 0$. $\sin x = 0$ gives 180° (the endpoints 0° and 360° are excluded); $\sin x = \frac{1}{2}$ gives 30° and 150° .

Question 5 TRG-15 · Trigonometric functions and equations · spec MM4.4

The graph of $y = \sin x$ (with x in radians) crosses the line $y = x$ at how many points?

Answer: D

$|\sin x| < |x|$ for all $x \neq 0$, so the only intersection is at the origin.

Question 6 TRG-17 · Trigonometric functions and equations · spec MM4.5

Which of the following is/are true for all real x ?

I $\sin^2 x + \cos^2 x = 1$

II $\sin(2x) = 2\sin x$

III $\sin(x + \pi) = -\sin x$

Answer: F

I is the Pythagorean identity. II is false (e.g. $x = \frac{\pi}{2}$: $0 \neq 2$). III: translating $\sin$ by half a period negates it, so III is true.

Question 7 TRG-10 · Trigonometric functions and equations · spec MM4.6

How many solutions does the equation $\sin x = \frac{x}{10}$ have?

Answer: A

Solutions need $|x| \leq 10$. Both sides are odd functions, so count $x > 0$ and double, then add $x = 0$. On $(0, \pi)$ the curve starts above the line (since $\sin x \approx x$) and ends below it: one crossing. On $(\pi, 2\pi)$, $\sin x < 0$: none. On $(2\pi, 3\pi)$ the line is between 0.63 and 0.94 while $\sin x$ rises to 1: two crossings ($3\pi \approx 9.42 < 10$). Beyond $4\pi > 10$ the line exceeds 1. So 3 positive, 3 negative and 0: seven solutions.